Advanced Programming Practice · 21CSC203P

Unit 5 · Session 4 · SLO 2 — Solved Worksheet

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Unit â€“ 5 : Session â€“ 4 : SLO â€“ 2 | Symbolic Expressions and Algebraic Structures

I. Questions & Answers

1. Symbolic expression â€” a mathematical expression built from symbolic variables (declared with sympy.symbols) that is stored unevaluated, e.g. x, y = symbols('x y'); expr = x**2 + 2*y.

2. Algebraic structure â€” a set equipped with one or more operations satisfying specific axioms, e.g. a group (closure, associativity, identity, inverse), a ring (addition + multiplication), or a field (a ring with multiplicative inverses).

3. Expanding (x+1)Â² in SymPy â€” expand((x + 1)**2) â†’ x**2 + 2*x + 1.

II. Code

- from sympy import symbols, expand
- x = symbols('x')
- print(expand((x + 1)**2)) # x**2 + 2*x + 1